

Syllabus

Quantitative Techniques for Financial Economics

EBC4097

Quantitative Techniques for Financial Economics

Academic Year: 2026-2027

Course Period: 1

SBE Emergency Number: 1333

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School of Business and Economics

📌 Academic year 2025-2026 Maastricht University School of Business and Economics

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Introduction

Summary: The objectives of the course are to provide student in the Financial Economics master program with a solid knowledge of stochastic models and econometric techniques used in the analysis of financial markets. The students should be able to read and assess the current literature on stochastic models and econometric methods used in security pricing and empirical finance and to apply the acquired techniques in practice. Topics that are covered include stationary and non-stationary time series models, volatility models and high-dimensional (Big Data) estimation methods. Empirical applications will provide students with practical experience in analyzing financial time series.

Prerequisites: Solid background in finance and in statistics/econometrics (on the level of a quantitatively oriented economics/finance bachelor). Required concepts from mathematics/statistics are random variables, probability distributions, statistical tests, regression analysis, ordinary and partial derivatives, exponential function. Basic knowledge in programming (e.g. in R, MATLAB, or similar) is suggested. Introductory material for R will be provided on the course web-page before the start of the course.

Related programs and specialisations: This course is a compulsory course for Master Financial Economics - Asset Pricing, Banking, Financial Analysis and no specialisation options. This course is a master exchange course for SBE Exchange Master and SBE Non-degree Courses.

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Course Objectives

Learning objectives of the course are the following:

- Evaluating the applicability of different time series models for a given financial problem.
- Translating financial research or application problems to canonical time series models.
- Applying time series models for describing and summarizing financial data.
- Applying time series models for evaluating the potential future outcomes in a financial problem.
- Interpreting and communicating the numerical results of time series models in a finance context.
- Collaborating to acquire necessary knowledge and skills.

This course aims to develop several competences including:

- Ability to translate a problem from practice or academia into a feasible research question, and to develop a strategy to address this research question.

- Ability to enrich an academic discussion of a practical problem.

Course objectives are related to the program objectives through:

- Providing essential knowledge in time series analysis, particularly taking the diverse student background into account.
- Understanding main topics, methods and applications as preparation to the following courses & thesis.
- Introducing main topics, methods and applications to help you choose following elective courses.

Course Structure

The course consists of two meetings per week.

- Attendance is compulsory for all tutorial meetings. You can miss up to **three tutorial meetings** throughout the course.
- Prior to the tutorial meetings, you are expected to work on a specific assignment in groups and prepare your answers. We then discuss the assignment within the tutorial. Each group is expected to actively participate in each tutorial.
- Each group is expected to have their answers ready, be able to share a screen and show their results to the group during these meetings. We suggest using MS Word or PDF files. Computer code should be presented only if it is strictly necessary.
- As we follow the PBL system in the tutorials, it is your responsibility to be well prepared to get relevant feedback from your colleagues and the tutor during these meetings.

Final group assignment

- During the course, you will work in groups on a final empirical assignment in financial economics. The assignment consists of a research proposal, a final empirical report and a final group presentation with individual questions. See the **Assessment and Grading** section below for details.
- The research proposal is intended to help you formulate a meaningful and feasible research question and develop an appropriate empirical strategy for addressing it.
- On **24 September**, each group will present its research proposal during the tutorial meeting. The presentation should briefly explain the research question, its economic motivation, the data to be used, the proposed econometric methodology and the main econometric or identification concern. The presentation is intended to provide an opportunity for discussion and feedback before the final empirical analysis is completed.
- The final assignment is due on **15 October at 23:00 hours**.
- At the end of the course, each group will present its final project. Each group will have **15 minutes** for the presentation and discussion. The final presentation should focus on the research question, economic motivation, empirical methodology, main results and their economic interpretation.

- Following the final group presentation, each student will be asked **one or two individual questions** about the project. Questions may concern any aspect of the research question, data, econometric methodology, results or economic interpretation. Every student is expected to understand the complete group analysis, regardless of their specific contribution to the project.

Assignments

The course consists of two types of assignments: tutorial assignments and final assignment.

Tutorial assignments

You do not have to submit your answers to these assignments. Instead you are expected to present your answers within the tutorial meetings.

Final assignment

- This is a final empirical group assignment in financial economics.
- It consists of a research proposal, a final empirical report and a final group presentation with individual questions.

Schedule and Deadlines

Meetings

The summary of meeting dates, topics and material are provided below. Resources for the meetings are briefly summarized and further resources will be given in the assignments.

Lectures & Tutorials

Tutorial assignments will be included prior to meetings in the course page.

Preparation material

The preparation material for the lectures and tutorials (pdf) will be posted in the course page. The R sessions, the preparation materials are provided in the link:

https://stephansmeekes.com/R_training/

Deadlines

- **24 September:** group research proposal presentations
- **15 October:** group final assignment presentations + individual questions
- **15 October:** group final assignment submission via the course page

Date	Meeting type	Topic	Preparation material
Tue, Sep 1, 26	L	Introduction lecture	EBC4097_Introduction.pdf
Thu, Sep 3, 26	T	R programming – basics	R training session 1, 2, 3, 4
Tue, Sep 8, 26	T	ARMA models	EBC4097_ARMA.pdf, EBC4097_UnitRoots.pdf
Thu, Sep 10, 26	T	Unit root testing	
Tue, Sep 15, 26	L	Multivariate time series models – VAR	EBC4097_VAR.pdf
Thu, Sep 17, 26	T	Multivariate time series models – VAR	
Tue, Sep 22, 26	T	R programming – intermediate	R training session 5
Thu, Sep 24, 26	T	Group assignment - research proposal presentation	Self and group study
Tue, Sep 29, 26	L	Conditional volatility models	EBC4097_ConditionalVolatility.pdf
Thu, Oct 1, 26	T	Conditional volatility models	
Tue, Oct 6, 26	L	Penalized regression	EBC4097_PenalizedRegression.pdf
Thu, Oct 8, 26	T	Penalized regression	
Tue, Oct 13, 26	T	Principal components analysis	EBC4097_PCA.pdf
Thu, Oct 15, 26	T	Group assignment – final presentation	Self and group study

Software Resources: R & R Studio

To prepare for the tutorials, you will need basic R programming skills. Basic time series models we cover are available in statistical software such as Eviews or SPSS. For most advanced models, the above statistical software is limited, so for this course we will use **R** programming language. You are free to choose any programming language, but support and code examples will be given from **R**. You will not be graded based on **R** code. The purpose in this course is for you to be able to do simple data manipulations and use existing code in **R** for time series analysis.

Throughout the tutorials, we will provide **R** code examples that you can manipulate and run for different tutorial assignments.

- To prepare for the meetings and assignments, you will need to download and install R and R Studio using: <https://www.r-project.org/> or <https://cloud.r-project.org/> for Windows, Mac and Linux.
- We also recommend to install R Studio desktop free for Windows, Mac and Linux: <https://www.rstudio.com/products/rstudio/download/>
- Self study the first 3 sessions in the R training link, which is provided as the preparation material for Tutorial 1: https://stephansmEEKES.com/R_training/

Assessment and Grading

Grading will be based on participation (20%), a final group assignment (30%) and a written exam (50%). A passing grade (5.5/10) is required from each assessment method. In case the student fails the written exam, a re-sit written examination is followed. In case the student fails participation or the final group assignment, a resit examination in the form of an individual course assignment will be provided.

Attendance to all tutorial meetings is compulsory. You can miss a **maximum of 3 tutorial meetings** in order to have a passing participation grade. Following the SBE policy, all education is on campus. So there will not be a Zoom link to join the meetings online.

The detailed grading scheme is provided in the table below.

Assessment Type	Grade type	Weight	Passing Condition	Details
Participation	Individual	20%	> 5.5 / 10	Evaluated during tutorials
Final assignment	Group	30%	> 5.5 / 10	Consists of a research proposal (5%), a final empirical report (15%), and a final group presentation with individual questions (10%)
Written exam	Individual	50%	> 5.5 / 10	Evaluated according to the submitted written exam

Participation assessment

The participation grade refers to the active involvement in tutorial meetings. In the tutorial meetings, groups of students will be assigned a specific task or questions related to the course material.

1. To obtain a sufficient grade (5.5/10), the students may not be absent in more than three tutorial meetings, demonstrate adequate preparation for the assigned group task, provide a presentation of the task and contribute to the in class discussions in tutorial meetings.
2. To obtain a good grade (6-7/10), in addition to point (1), the student should demonstrate a concise presentation of their case study including a detailed relation to the overall course topic as well as a thorough interpretation of the group's results and findings.
3. To obtain a very good grade (8-9/10), in addition to points (1&2), the student should actively participate in the in class discussions during other groups' presentations with consistent involvement in the discussions, responding to other students' questions and presentations and questioning the case contents in an academic and constructive manner.
4. To obtain an excellent grade (10/10), in addition to points (1&3), the student should prepare an exceptionally good case presentation and discussion which goes beyond the assigned task by relating the content and case to other existing approaches, including a discussion on relevant

and timely possibilities of further development of the case. In addition, the students should contribute in a very substantial way to the ongoing discussions in all group meetings and improve the learning process for the whole group by providing constructive arguments and clarifications to other students' comments and questions.

Final assignment (30%)

The final assignment is a group-specific empirical project in which you will apply the methods and techniques acquired during the course to a meaningful question in financial economics. The assignment consists of a research proposal, a final empirical report and a final group presentation with individual questions.

Research proposal (5%)

Each group submits a short research proposal at the beginning of the project. The purpose of the proposal is to formulate a meaningful and feasible research question and to develop an appropriate empirical strategy for addressing it.

The proposal should include:

- the research question and its economic motivation;
- the main hypothesis or hypotheses;
- a description of the data to be used;
- the proposed econometric methodology;
- the main econometric or identification concern; and
- a brief explanation of what the analysis is intended to contribute to our understanding of the question.

The research proposal will be assessed based on the quality and relevance of the research question, the economic reasoning, the feasibility of the proposed analysis, and the appropriateness of the proposed econometric methodology.

Final empirical report (15%)

The final report should present the empirical analysis conducted by the group. The report should be **6–8 pages**, excluding references and appendices. The report should be structured around the following elements:

1. **Research question and economic motivation:** Clearly state the question and explain why it is meaningful in financial economics.
2. **Data:** Describe the data, variables and relevant data construction choices.
3. **Empirical methodology:** Explain and motivate the econometric methods used.
4. **Results:** Present the main empirical findings using appropriate tables and figures.
5. **Economic interpretation:** Explain the magnitude and economic significance of the findings, in addition to their statistical significance.

6. **Robustness and limitations:** Discuss the main limitations of the analysis and, where appropriate, provide robustness checks.
7. **Conclusion:** Summarize the main findings and what can and cannot be concluded from the analysis.
8. **Division of labour:** Briefly describe the contribution of each group member to the project.

The grading of the final report will focus primarily on **economic reasoning and the correctness and appropriateness of the econometric analysis**. A simple empirical method that is appropriately chosen and correctly applied is preferable to a more sophisticated method that is not appropriate for the research question.

The group should submit the report together with the relevant computer code used to conduct the empirical analysis. Each group member is expected to understand the complete analysis, regardless of their specific contribution to the project.

Final presentation and individual questions (10%)

At the end of the course, each group will present its final project. Each group will have **15 minutes** for the presentation and discussion.

The presentation should focus on the research question, the economic motivation, the empirical methodology, the main results and their economic interpretation. The presentation should not attempt to reproduce the entire report, but should communicate the main elements and findings of the analysis clearly and concisely.

Following the group presentation, each student will be asked **one or two individual questions** about the project. Questions may concern any aspect of the research question, data, econometric methodology, results or economic interpretation. Every student is therefore expected to understand the complete group analysis, and not only the part for which they were primarily responsible.

The individual questions are intended to assess whether each student understands and can explain the group's empirical analysis and the economic reasoning underlying it. Additionally, there is a penalty for mistakes in submission such as submitting a wrong file format or missing team information.

Written examination

The written examination is arranged on campus. Please check your timetables for the date and place of this exam. This examination consists of open questions. An example examination and submission format for your answers will be provided in the course page.

Validity of partial results

The rule from the EER is the following: Parts of an educational unit which were passed will remain valid in the academic year in which the partial results are obtained and two (2) more academic years, without prejudice to the competency of the Board of Examiners to extend this period of validity. In case the course changes substantially from the previous year, transitional

rules for repeat students need to be clearly indicated in the course manual. Also ensure to check the distinction between 'exam only' registrations and regular registrations.

Appendices

Possible appendices:

- Feedback forms
- Specific requirements course assignments, papers etc., for example the correct form of citation and references
- Tips and tricks for skills you are teaching your students (e.g. presentation skills, academic writing etc.)

Student Code of Conduct

How do we wish to interact with each other? Here you will find a highlight of the values of the



School of Business and Economics (SBE), which inspire our community and act as a framework for the personal conduct of employees and students:

01 Respect

First of all, we should be able to count on each other: among staff, among students, and in cooperation with each other. Respect, reliability, taking responsibility and self-reflection are the core concepts here. This includes, but is not limited to, the use of appropriate language in both written and spoken communication.

02 Commitment

We are part of a community of motivated students and staff. We expect students and staff to express their intellectual passion through ambition, initiative and commitment. Staff and students must encourage each other in this process.

03 Professionalism

Staff and students contribute to an inspiring work and academic climate from their various fields of expertise, knowledge and experience. We value each other's contribution in word and gesture. Students and employees have many rights as well as obligations and we acknowledge these and apply them appropriately and professionally at all times.

04 Inclusivity

Our School is dedicated to creating an inclusive environment for everyone, regardless of race,

ethnicity, religion, colour, nationality, sexual orientation, gender etc. The inclusion of all students and staff members is fundamental to our School's diverse and international character. English is used as the lingua franca at all times within our School.

05 Integrity

We are committed to protecting and guaranteeing academic integrity. This means taking exams independently and honestly, indicating sources when writing a paper, essay or thesis and always being truthful when filling out forms and other documentation.

This Code of Conduct offers the possibility to hold someone accountable for undesirable behaviour. For more details, including complaint procedures and potential sanctions, please consult our full code of conduct on SBE's Student Intranet.

Please include the following appendices for legal reasons:

Fraud and Plagiarism

In order to protect the reputation of the degrees that you – as students – receive, instances of cheating or plagiarism are treated extremely seriously.

Fraud, including plagiarism, is understood as a student's act or failure to act that makes it partially or fully impossible to correctly assess his/her knowledge, insight and skills. Plagiarism is understood as the presentation of one's own or other people's ideas or words without adequate reference to the source. Any assignment is an individual piece of work, which means that plagiarism is strictly forbidden. Equally, the use of mobile phones, communication devices or any other information carrier (whether the phone or other device is turned on or off, used or not used, etc. is irrelevant) during an examination is also forbidden.

If the Board of Examiners concludes that anything has occurred in an examination that makes it partially or fully impossible to correctly assess his/her knowledge, insight and skills, they may impose a sanction in accordance with SBE's policy on fraud, including plagiarism.

More information can be found on MySBE Intranet:

<https://intranet.maastrichtuniversity.nl/en/cheating-and-plagiarism>

Partial results

If students fail a course and have to redo it in the next academic year, they have two options:

1. Register for the exam only (assuming the typical case in which a student has failed the written exam but has passed other parts of the overall examination). Other partial grades remain valid and the final grade will be calculated using the weights that were used in the academic year in which the student followed the course.

2. Register for education. If a student decides to register for tutorials, (s)he will get new grades for those grades that are assessed during the tutorial meetings (e.g. participation, presentation). This makes sure that students that register for tutorials can't passively 'consume education', but instead are required to actively participate in all educational activities

Resits

Make clear that according to EER, exclusion is only permitted if:

The exclusion of a resit opportunity for (a) component(s) of (the examination of) an educational unit is only permitted if the aforementioned component(s), because of its/their nature can only be assessed within the relevant course period (e.g. participation, software labs, presentation). The reason for the exclusion of a resit should be explained and motivated in the course manual.

Comments, objections and appeal

Students can find detailed information on these procedures on the Intranet:

<https://intranet.maastrichtuniversity.nl/en/comments-complaints-and-appeal> as well as the EER: <https://intranet.maastrichtuniversity.nl/en/education-and-examination-regulations>

1. Comments

General) Comments regarding written examinations with open and/or closed questions (Multiple Choice) or parts thereof have to be submitted to the course coordinator according to the rules as published on the Intranet via <https://examcomments.sbe.maastrichtuniversity.nl/>. Deadline is 24:00 of the fifth day after the exam.

2. Inspection and objections

Within ten (10) working days of the publication of your examination results in the Student Portal >> My Courses >> Show results, you will be able to have a look at your assessed work (exam inspection).

You can request a copy of your multiple choice answer sheet via Surfyourself (SYS) in order to inspect your examination.

It is your course coordinator's responsibility to announce when and how you can do this. Usually, the date(s) will be posted with the (digital) publication of examination results or on the course page in My Courses.

If you can demonstrate that you were not able to make a complaint within the specified period due to circumstances beyond your control, you will be given another ten (10) working days period in which to do so.

If you do not agree with your assessment, you can submit objections in writing during the inspection, according to the instructions communicated by the course coordinator.

3. Complaints and appeal

You can send a complaint (not being a comment or an objection as mentioned above) regarding the course to the Complaints Service Point CSP:

<https://www.maastrichtuniversity.nl/support/during-your-studies/complaints-service-point-csp-one-service-desk-objections-appeals-or-0>

If you and the course coordinator cannot reach an agreement or you are not satisfied with a decision of an examiner or the Board of Examiners, you can lodge an appeal to the Board of Appeal for Examinations (College van Beroep voor de Examens, CBE), within six (6) weeks after you received the decision. You may only appeal if a decision has gone directly against your interests. More information about what you may lodge an appeal against with the Board of Appeal for Examinations can be found on the Intranet and at the CSP link above.